

Neural Networks Field Guide

The four architectures, when to use which, and what to ask before trusting any AI system.

ANN Artificial Neural Network

The general-purpose original: layers of simple neurons, each layer refining the last.

Reach for it when: many variables combine in messy ways. Credit scoring, default prediction, customer churn.

CNN Convolutional Neural Network

Built for images. Early layers find edges, later layers assemble shapes and objects.

Reach for it when: the data is visual. Document processing, check and invoice reading, fraud-pattern recognition.

RNN Recurrent Neural Network

A network with memory. Reads sequences step by step and carries context forward (LSTM extends the memory).

Reach for it when: order and history are the signal. Time-series forecasting, demand planning, cash-flow projection.

GAN Generative Adversarial Network

Two networks in a contest: a generator makes fakes, a discriminator judges them, both improve.

Reach for it when: you need realistic synthetic data for testing without exposing real records. Also the engine behind deepfakes: treat outputs with care.

CHOOSING IN ONE LINE

Tables of numbers: **ANN**

Pictures or documents: **CNN**

Anything with a timeline: **RNN / LSTM**

Creating realistic data: **GAN**

THREE QUESTIONS TO ASK BEFORE TRUSTING ANY AI SYSTEM

1. What was it tested against? Ask for the adversarial test set and the refusal or error rate, not a demo. Guardrails that were never attacked are assumptions, not guardrails.
2. Is it consistent? Send the same request many times. A system that answers differently every time cannot anchor a process, no matter how impressive one answer looks.
3. How is success measured with real users? Task completion and user trust should be tracked over time. If nobody can show you the numbers, trust is being asked for, not earned.

Why measurement? Deep networks are black boxes: inputs and outputs are visible, the internal reasoning is not. Trust has to come from observed behavior, verified repeatedly, not from inspection.